MEDICO ENGINEER

MEDICO ENGINEER

Your Free MDCAT Companion

Serving MDCAT students for 4 years — completely free of cost

Prepare smarter with trusted, high-quality MDCAT resources designed to help students practice, revise, and improve confidently.

We Provide

Free High-Quality Practice Tests

Chapter-wise and topic-wise MDCAT practice tests for strong exam preparation.

MDCAT Books

Useful books and study resources for Biology, Chemistry, Physics, English, and Logical Reasoning.

MDCAT Material

Organized MDCAT preparation material for students.

Notes & Revision Guides

Easy notes for quick revision and concept clarity.

Top Academies Material

Selected material from well-known academies to support your MDCAT preparation.

Access Free Testing Service

Join Our WhatsApp Channel

Scan the QR Code to Join

Connect With Us

Facebook Page: Medico Engineer

WhatsApp: 03700113837

Website: Medicoengineer.com

Medico Engineer

Free MDCAT Preparation Platform for Every Student

Prepare • Practice • Improve • Succeed

SINDH

MDCAT PAST PAPER

2020

FOR MORE INFORMATION VISIT OUR FACEBOOK PAGE
PRESTIGIOUS TESTING SERVICE-PTS

COMPILED BY: RAM LAL VALASAI

Medicoengineer.com

BIOLOGY

1. When the temperature of the body surrounding rises, baby responds by:
A. Vasoconstriction
B. Vasodilation
C. Shivering
D. Raising body hairs

2. The excretion of hypertonic urine in humans is associated best with the:
A. Glomerular capsule
B. Proximal convoluted tubule
C. Loop of henle
D.Distal convoluted tubule

3. In humans, the temperature regulation control center is located in:
A.Kidneys
B. Brain
C. Lungs
D.Liver

4. As an excretory organ, liver:
A.Detoxifies many chemical poisons
B. Produces ammonia for excretion by the kidneys
C. Produces urine and uric nitrogen of amino acids
D.All of above

5. The active uptake of sodium in the ascending limb or thick loop of henle is promoted by the action of:
A.Aldosterone
B. Thyroxine
C. ADH
D.Cortisone

6. Which of the following muscles are considered as 'Voluntary Muscles" ?
A.Smooth muscles
B. Cardiac muscles
C. Skeletal muscles
D.Glandular muscles

7. Which types one of the following "myogenic" type of muscle?
A.Glandular muscles
B. Cardiac muscles
C. Skeletal muscles
D.Smooth muscles

8. What do we call the cell surface membrane of a muscle fiber?
A.Sarcolemma
B. Plasma membrane
C. Sarcoplasm
D.Myofibrils

9. Which of the following neurotransmitters function, both neurotransmitter and hormones, decreasing our perception of pain?
A.Epinephrine
B. Serotonin
C. Dopamine
D.Endorphins

10. Which body function is controlled through a positive feedback mechanism?
A.Labor contractions
B. Body Temperature
C. Insulin production
D.Thyroxine release

11. Which of the following is common to all neurons?
A.A cell which contains a nucleus
B. A thick myelin sheath
C. Presence of node of Ranvier
D.Presence of Schwann cells

12. Neurons are cells adapted for the rapid transmission of electrical impulses.
To do this, they have long thin process called:
A.Axons
B. Dendrites

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023
Medicoengineer.com

C. Myelin sheath
D.Schwann cells

13. _____ is a junction between two neurons or between a motor neuron and a muscle cell:
A.Impulse
B. Synapse
C. Axon
D.Cleft

14. Which of the following represents the changes that occur in the ovary and the uterus approximately every 28 days involving ovulation with the breakdown and loss of the lining of the uterus?
A.Ovulation
B. Menstrual cycle
C. Uterine Cycle
D.Embryo formation

15. Which of f the following diseases Is sexually transmitted?
A.Tuberculosis
B. AIDS
C. Dengue fever
D.Cholera

16. Which of the following hormones of the pituitary gland regulate the menstrual cycle?
A.Follicle stimulating hormone and estrogen
B. Luteinizing hormone and estrogen
C. Follicle stimulating hormone and luteinizing hormone
D.Estrogen and progesterone

17. Haemophilia A and B , colour blindness and testicular feminization are
example of:
A.X - linked dominant trait
B. Y - linked recessive trait
C. Y - linked inheritance
D. Pseudoautosomal trait
E. None of these

18. Which traits are most likely to affect men than women?
A. The form of appearance of a trait
B. X linked dominant
C. Autosomal dominant
D. Autosomal recessive

19. Alleles both have an effect on the phenotype heterozygotic organism:
A.Dominant
B. Recessive
C. Multiple
D.Co-dominant

20. When both the allele of a genes are same, the organism is said to be:
A.Heterozygous
B. Genotype
C. Homozygous
D.Phenotype

21. In which type of cell, Cell wall is not present?
A.Plant cell
B. Fungal cells
C. Bacterial cells
D.Liver cells

22. 70S size ribosomes are found in the cells of:
A.Algae
B. Protozoans
C. Fungi
D.Bacteria

23. According to the fluid mosaic model of cell membrane, which zone is embedded inside?
A.Hydrophobic
B. Globular
C. Hydrophilic
D.Filamentous

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

24. The membrane separating the vacuole from cytoplasm is called:
A.Cristae
B. Cisternae
C. Tonoplast
D.Vacuolar membrane

25. Select the one which is not a function of smooth endoplasmic reticulum (SER) ?
A.Metabolism of lipids
B. Transmission of impulses
C. Transport of materials
D.Processing of glycoproteins

26. Which of the following organelles are involved in the synthesis of plant cell wall?
A.Endoplasmic reticulum
B. Golgi complex
C. Lysosomes
D.Peroxisomes

27. Which property of water helps to maintain the integrity of cell membranes?
A.Specific heat capacity
B. Hydrogen bonding
C. Cohesion and adhesion
D.Hydrophobic exclusion

28. Water act as universal solvent because of:
A.Heat of vaporization
B. Hydrogen bonding
C. High polarity
D.Cohesion and adhesion

29. Lipids store double amount of energy as compared to carbohydrates because of:
A.High proportion of oxygen
B. High C-O ratio
C. Low proportion of carbon
D.High proportion of C-H

30. Which of the following is an unsaturated fatty acid?
A.Oleic acid
B. Palmitic acid
C. Butyric acid
D.Acetic acid

31. Monosaccharides have a general formula represented by:
A.Cn(H2O)n
B. C(H2O)n
C. C2(HO)n
D.C2(H2O)n

32. NAD is an example of:
A.Mononucleotide
B. Dinucleotide
C. Tri nucleotide
D.Tetra nucleotide

33. Lock and key model of enzyme action proposed by Emil Fischer suggested that:
A.Enzymes are unbiased for the substrate
B. Enzymes are not restricted to one reaction type
C. Enzymes are restricted to one reaction type
D.An enzyme can catalyze variety of reactions

34. Most enzymes have an optimum temperature of around:
A.30°C
B. 40°C
C. 50°C
D.20°C

35. Enzymes work by lowering the _____of reactions that catalyze:
A.Kinetic energy

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

B. Activation energy
C. Heat energy
D.Potential energy

36. First stable compound during calvin cycle is:
A.3-phosphoglycerate
B. Glyceraldehyde 3-phosphatse
C. 1,3 bisphosphoglyceratae
D.Ribulose biphosphate

37. What is the function of ribulose?
A.Intermediate in photosynthesis
B. Respiratory fuel
C. Intermediate in cellular respiration
D.Component of DNA and RNA

38. Which of the following processes does not need pyruvic acid as a substrate?
A.Alochol fermentation
B. Calvin cycle
C. Aerobic respiration
D.Lactic acid fermentation

39. Which of the following is a copper containing protein in electron transport chain?
A.Plastoquinone
B. Cytochrome-C
C. Plastocyanin
D.Ferredoxin

40. In electron transport chain, ATP synthesis takes place when electron moves from:
A.Primary electron acceptor (PEA) to plastoquinone
B. Plastoquinone (Pq) to cytochromes
C. Cytochromes to plastocyanin
D.Plastocyanin (Pc) to photosystem I (PS-I)

41. "Law of independent assortment" states:
A.The each pair of alleles assort independent of other pairs of alleles during gamete formation
B. That allele for each pair of contrasting trait have unequal probability to assort with the alleles of other pairs
C. That coexisting alleles for each trait segregate (separate) from each other at meiosis so that each gamete receives only one of the two alleles
D. That pertain to inheritance of single trait (monohybrid cross)

42. Phenotype is:
A.The genetic complement i.e the genes in an individual for a particular trait
B. Partner of gene pair
C. The form of appearance of a trait
D.The position of a gene on the chromosome

43. In complete dominance:
A.Different alleles of a gene are both expressed in heterozygous condition
B. One allele (R) is completely dominant over the other (r) and the presence of recessive allele is functionally hidden the heterozygote (Rr) has the same sound phenotype (RR) heterozygote
C. The phenotype of the heterozygote is intermediate between phenotypes of the homozygotes.
D. Gene mutation may produce many different alleles of a gene

44. Which one of the following is found in both messenger RNA and DNA of a mammalian cell?
A.Double helical structure
B. Ribose Sugar
C. Thymine
D.Sugar- phosphate backbone

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

45. The cells in our body are all genetically identical apart from the:
A.Somatic cells
B. Reproductive cells
C. Muscle fibers
D.White blood cells

46. Transcription is the process in which an RNA copy of the DNA sequence and coding the gene is produced with help of an enzyme called:
A.DNA polymerases
B. RNA polymerase
C. DNA transcriptase
D.RNA transcriptase

47. The particular array of chromosomes that an individual posses is called its:
A.Genotype
B. Phenotype
C. Karyotype
D.Allele

48. During meiosis, the homologous chromosomes comes together and form pairs this process is called:
A.Linkage
B. Synapsis
C. Pairing
D.Crossing over

49. At what phase the DNA content of a cell is doubled?
A.Prophase
B. Interphase
C. Anaphase
D.Telophase

50. Which statement correctly describes transcription of DNA?
A.It produces amino acids
B. It produces messenger RNA
C. It results in increased DNA synthesis
D.It is a semi conservative process

51. This theory says that "mitochondria and chloroplast are infact ancient bacteria which now live inside the large cells":
A.Darwin's theory of evolution
B. Lamarckism
C. Neo-darwinism
D.Endosymbiont theory

52. The organs which are similar but differ in structure are called:
A.Analogous organs
B. Homologous organs.
C. Convergent evolution
D.Divergent evolution

53. _____ occurs because natural selection gives some alleles a better chance of survival than others:
A.Fitness
B. Evolution
C. Crossing over
D.Artificial selection

54. The DNA that has been altered and which now contains length of nucleotides from two different organisms is called a:
A.Plasmid
B. Combined DNA
C. Vector
D.Recombinant DNA

55. It is a method for rapid production of a very large number of copies of a particular fragment of DNA:
A.Gel electrophoresis
B. Polymerase chain reaction
C. DNA extraction
D.Recombination

56. What is the effect of DNA ligase?
A.DNA is broken up at specific sites

MDCAT Past Papers BY PRESTIGIOUS TESTING SERVICE 2000-2023

B. DNA fragments are joined together
C. DNA replication occurs
D.DNA transcription occurs

57. Which of the following is the components/tools of recombinant DNA technology?
A.Gene of interest
B. Molecular scissors
C. Molecular glue and expression system
D.All of the above

58. Gel electrophoresis is a technique:
A. Employed by forensic scientists to assist in the identification of the individuals by their respective type of DNA
B. Collect all the genes found in one complete set of chromosomes
C. Is a technique to separate different size fragment of charge bearing polymers (proteins, RNA or DNA)
D. Grows single cells of a group of cells in a glass ware on artificial medium under asepecific conditions.

59. Transgenic organisms:
A.Have a foreign gene inserted into them
B. Have an important role in the large scale production of medicinal products
C. Are considered beneficial to humans
D.All of the above

60. Which of the following is not necessary for PCR to occur?
A.dATP
B. Primers
C. DNA fragments
D.Ribonucleotides

61. The end product of glycolysis in aerobic respiration is:
A.Ethanol and carbon dioxide
B. Lactate
C. Pyruvate
D.Acetyl CoA

62. Which of the following is not related to enveloped virus?
A.They survive for a short time
B. Their envelop is sensitive to sunlight
C. They are tolerant to antidotes
D.Envelope is derived from host

63. Numerous opportunistic diseases might attack a person suffering from which of the following diseases?
A.Measles
B. Influenza
C. Hepatitis A
D.AIDS

64. The complete, mature and infectious virus particle is known as:
A.Venome
B. Genome
C. Virion
D.Capsid

65. Which of the following is NOT TRUE about Human Immunodeficiency virus (HIV)?
A.It is retrovirus
B. It is surrounded by an envelope
C. It does not cause AIDS
D.It causes the deficiency of the human immune system

66. Select a method which cause the oxidation of constituent of a bacterial cell:
A.Steam
B. Filtration
C. Dry heat
D.Radiation

67. Which of the following is TRUE about the structure of a typical bacterium?
A.It has a cell wall

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

B. It has cytoplasm
C. It has genetic material
D.All of the above

68. Red algae do not contribute towards:
A.Making coral reefs
B. Forming limestone deposits
C. Making fertilizers
D.Forming chalk deposits

69. Which of the following is true about amoeba?
A.They have flagella
B. They are multicellular
C. They do not cause any disease in humans
D.They move by forming specialized cytoplasmic projections called pseudopodia

70. The directional movement toward or away from the stimulus is called:
A.Tropism
B. Orientation
C. Taxis
D.Non orientation

71. Photophosphorylation takes place in the _____ of the chloroplasts.
A.Stroma
B. Granum
C. Inner membrane
D.Outer membrane

72. Select an amniote from the following:
A.Snake
B. Frog
C. Parrot
D.Crocodile

73. In roots the apoplast pathway of water is disrupted when water reaches:
A.Plasmodesmata
B. Cortex
C. Endodermis
D.Pith

74. Regarding structure of human heart chordae tendinea are present in:
A.Atria
B. Pulmonary valve
C. Ventricles
D.Aortic valve

75. The only vein in human body carrying oxygenated blood is:
A.Femoral
B. Pulmonary
C. Renal
D.Iliac

76. The cells which play very important in developing immunity are:
A.Monocytes
B. Neutrophils
C. Lymphocytes
D.Thrombocytes

77. Which of the following blood vessels have the highest pressure of blood?
A.Aorta
B. Pulmonary arteries
C. Pulmonary veins
D.Vena Cava

78. Autoimmune disease act at the principle of:
A.Self against antigens
B. Antigens against self
C. Self against self
D.Antigens self-destroyed

79. In human heart, the left atrium receives:
A.The superior Vena Cava
B. The inferior Vena Cava
C. The Coronary sinus
D.The four pulmonary veins

MDCAT Past Papers
PRESTIGIOUS TESTING SERVICE
2000-2023

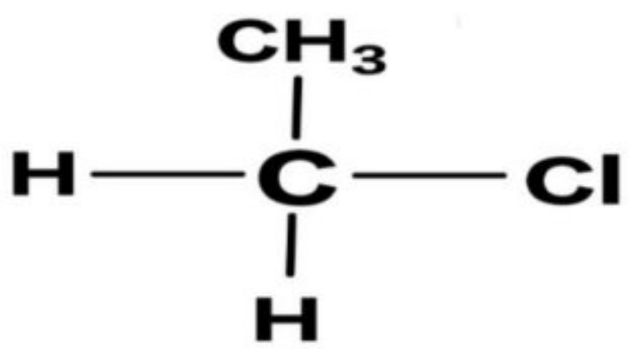
80. Antibodies are manufactured in:
A.T-lymphocytes
B. Red blood cells
C. Platelets
D.B-Lymphocytes

CHEMISTRY

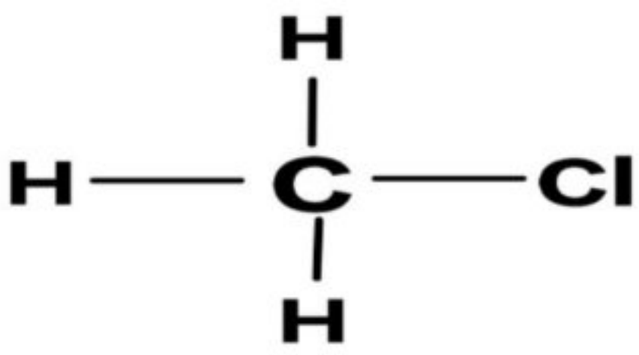
81. Alkyl halides involving _C_X bond breakage and _C_NU bond formation simultaneously would follow the mechanism:
A.SN1
B. SN2
C. E1
D.E2

82. Secondary alkyl halide is:

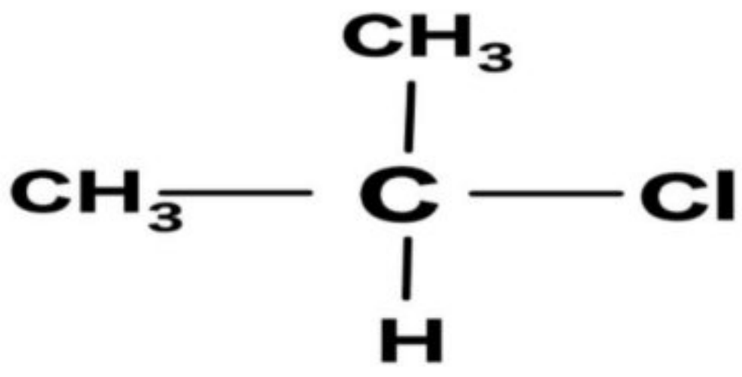
A.



B.



C.



D.

CH_3Cl

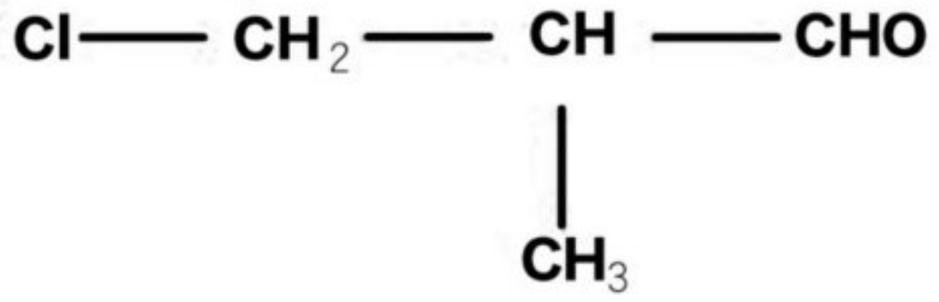
83. R-X reaction with alcohols forms:
A.R-OH
B. ROR
C. R-X-OH
D.RH

84. IUPAC name of C6H5O(CH3)2 is :
A.2-methyl-3-hexanone
B. 2,6-Dimethyl cyclohexanone
C. 3 methyl cyclohexanone
D.4-methyl-3-hexanon

85. Phenol is known as:
A.Carpolic acid
B. Carbonylic Acid
C. Carbolic Acid
D.Carboxylic Acid

86. Phenol is more acidic than alcohols because of the following reasons:
A.Delocalization of negative charge in the OH group
B. Delocalization of positive charge on the carbon atom in ring
C. Delocalization of negative charge in the ring
D.Delocalization of positive charge in the OH group

87. The common name of following aldehyde:



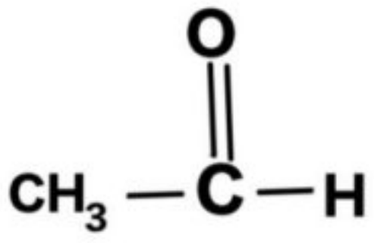
A.α-Methyl-γ-Chloro Propionaldehyde
B. β-Chloro-γ-methyl Propionaldehyde
C. β-Chloro-α-methyl Propionaldehyde
D.β-methyl-α-chloro Propionaldehyde

**88. Which of the following regent is used to separate purify carbonyl and non-
Carbonyl compounds?**

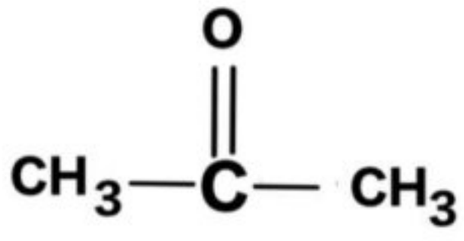
MDCAT Papers TESTING SERVICE

A.HCN
B. BrMgCH3
C. NaHSO3
D.H2O

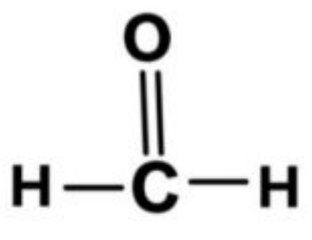
89. Secondary alcohol is the product of reduction of which carbonyl compound?



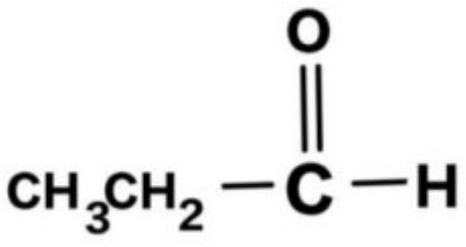
A.



B.



C.



D.

90. Which of the following is the strongest acid?
A.Propionic acid
B. Fluoroethanoic acid
C. Trichloroethanoic acid
D.Nitro Ethanoic acid

91. Hydrolysis of acyl chloride results in the formation of:
A.Acid anhydride
B. Carboxylic acid
C. Amides
D.Esters

92. The exact reactivity order for carboxylic acid derivatives is:
A.Anhydride > Acyl Chloride > ester
B. Ester > Anhydride > Acyl chloride
C. Amide > Acyl chloride > ester
D.Acyl Chloride > Anhydride > ester

93. Based on the physio-chemical properties, proteins may be classified into the following types:
A.Simple proteins
B. Compound proteins
C. Derived proteins
D.All of the above

94. Based on function, thyroxin can be classified as:
A.Hormonal protein
B. Structural protein
C. Transport protein
D.Genetic protein

95. L-Asparaginase enzyme has been used for the treatment of:
A.Jaundice
B. Blood cancer
C. Rickets
D.Heart disease

96. Potassium, Rubidium, Cesium react with oxygen to form which types of oxides?
A.Peroxide
B. Superoxide
C. Suboxide
D.Normal Oxide

97. Magnesium reacts with Nitrogen to form:
A.Mg2N2
B. Mg3N2
C. MgN2
D.MgN

98. Densities of alkali metals are low due to:
A.Weak intermolecular forces
B. Large atomic volume
C. Smaller size
D.ns1 configuration

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

99. In 3rd series of transition elements paramagnetic behaviour is maximum for Mn+2 and:
A.Cr3+
B. Ti3+
C. V3+
D.Zn+2

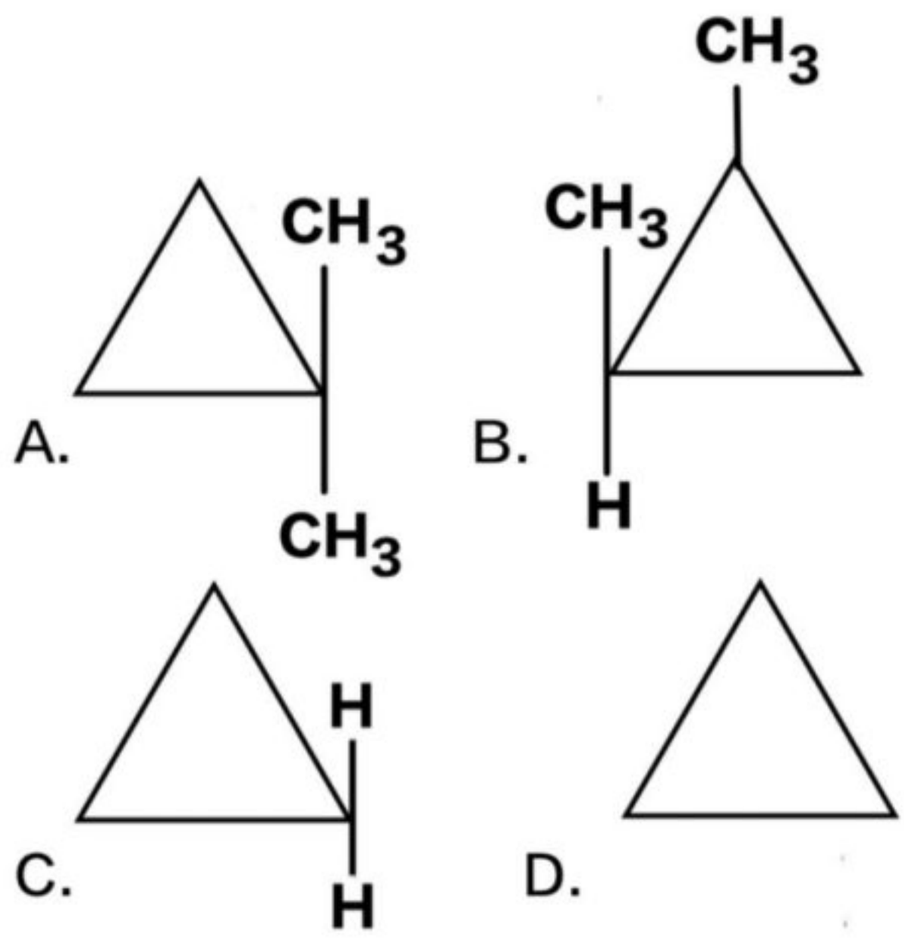
100. Electronic configuration of chromium (Proton number 24) is:
A.[Ar]3d44s2
B. [Ar]3d54s2
C. [Ar]3d54s1
D.[Ar]3d64s2

101. The transition element which does not show variable valency is:
A.Cu
B. Sc
C. Zn
D.Cr

102. Select the organic compound which belongs to Arene family:
A.CH2=CH2
B. CH3-O-CH3
C. CH3-NH2
D.C6H6

103. The type of isomerism existing in a compound of molecular formula C2H6O is:
A.Functional group
B. Position
C. Chain
D.Metamerism

104. Which of the following compounds which show geometric isomerism?

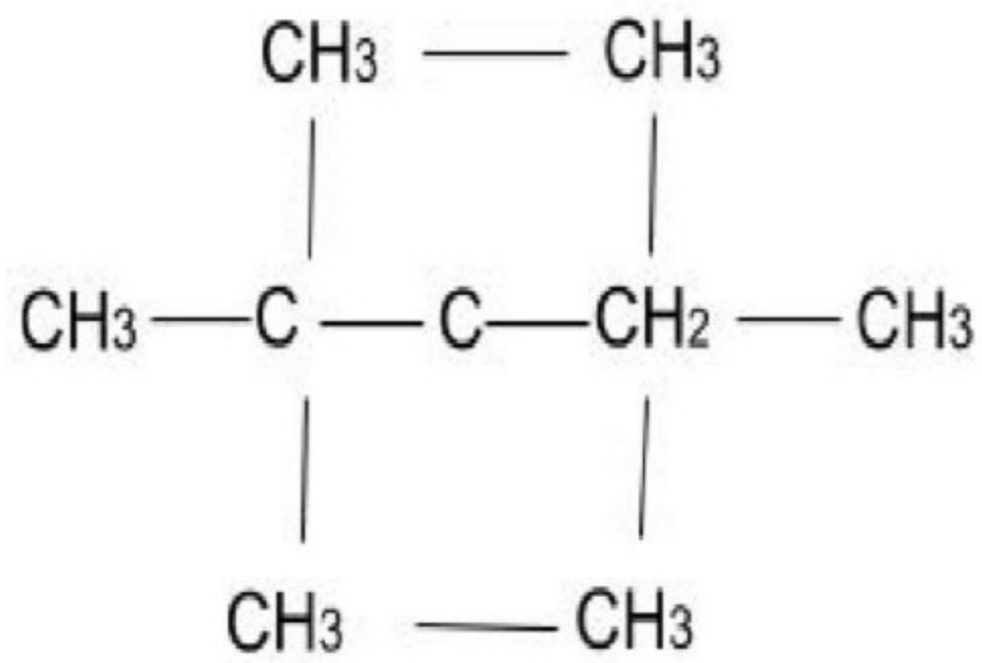


A.A
B. B
C. C
D.D

105. General formula of cycloalkane is?
A.CnH2n+2
B. CnH2n
C. CnH2n+1
D.CnH2n-2

106. Electrophile in sulphonation of benzene is:
A.HSO4-
B. H2SO4
C. SO3
D.HSO3-

107. The following has IUPAC name of:



A.2,3-tetramethyl butane

MDCAT
Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000–2023

B. 2,2,3,3-tetramethyl pentane
C. 3,3,4,4-tetramethylbutane
D.3,4-bis (dimethyl butane)

108. Acetophenone can be formed by which of the following reaction of benzene?
A.Alkylation
B. Acylation
C. Halogenation
D.Nitration

109. In alkanes, each Carbon has hybridization:
A.sp^3
B. sp
C. sp^2
D.dsp

110. When CH3 is attached with the benzene ring, it makes the ring:
A.Good electrophile
B. Good nucleophile
C. Resonance hybrid
D.Extraordinary stable

111. Which of the following reaction has greater Kp than Kc (Kp > Kc)?
A.$2NO + CL2 \rightleftharpoons 2NOCL$
B. $2SO2 + O2 \rightleftharpoons 2SO3$
C. $2NOCl \rightleftharpoons 2NO + CL2$
D.$NO2 + 3H2 \rightleftharpoons 2NH3$

112. The equation N2g+ 3H2g = 2NH3, represents:
A.Contact process
B. Haber's process
C. Solvay proces
D.Avogadro's law

113. The unit of the rate constant is the same as that of the rate of reaction in :
A.Zero order reaction
B. First order reaction
C. Second order reaction
D.Third order reaction

114. The study of rates of chemical reactions and the factors that affect the rates of chemical reactions is known as:
A.Thermodynamics
B. Stoichiometry
C. Electrochemistry
D.Chemical kinetics

115. For the reaction A(g) → products, When the concentration of A(g) 'doubles,
the rate of reaction increases four folds, which means it is:
A.Negative order reaction
B. First order reaction
C. Zero order reaction
D.Second order reaction

116. For which of the following order of the reaction, rate of reaction in inversely proportional to the concentration of reaction?
A.1st order reaction
B. 2nd order reaction
C. Negative order of reaction
D.Zero order of reaction

117. The thermal energy at constant pressure is called:
A.Enthalpy
B. Internal energy
C. Heat capacity
D.Work done

118. Born-Haber cycle is used to determine the lattice energies of:
A.Molecular solids
B. Metallic solids
C. Ionic solids
D.Covalent solids

119. One calorie is equal to:
A.4.18 KJ

MDCAT Past Papers
PRESTIGIOUS TESTING SERVICE
2000-2023

B. 4.18 J
C. 0.418 KJ mol
D.0.418 KJ

120. The oxidation state of 'S' in the (S2O3)-2 is:
A.+4
B. +6
C. -2
D.+2

121. The common oxidation number of halogens is:
A.-1
B. +1
C. -2
D.0

122. During oxidation process, oxidation number of an element:
A.Decreases
B. Increases
C. Remains constant
D.Both A & B

123. Which of the following has the highest value of electronegativity?
A.I
B. Br
C. Cl
D.F

124. Which of the following hybrid orbitals has maximum 'S'-character?
A.sp^3 -hybrid orbital
B. sp^2 -hybrid orbital
C. sp-hybrid orbital
D.dsp^2 -hybrid orbital

125. The first ionization energy is maximum for:
A.Na
B. Mg
C. Al
D.K

126. The efficiency of chemical reaction can be expressed as:
A.Theoretical yield
B. Actual yeild
C. % yield
D.Maximum yeild

127. In a vessel, 10g N2, 10g H and 10g O2 are present. Which one will have least number of atoms?
A.H2
B. N2
C. O2
D.Both N2 & H2

128. The empirical formula of Glucose C6H12O6 is:
A.C6H12O16
B. CHO
C. CH2O
D.CH2O2

129. The relationship between quantum number n and l is:
A.n = l - 1
B. l = n - 2
C. l = n - 1
D.n = l - 2

130. Quantum number values for '2p' orbitals are:
A.n = 2 , l = 1
B. n = 1 , l = 2
C. n = 1 , l = 0
D.n = 2 , l = 0

131. Which pair has 1 electron in it's 's' orbital?
A.Li , Fe
B. Na & Cr
C. K & Mn
D.H & He

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

132. Which of the following has the lowest e/m ratio?
A.Li+2
B. H+1
C. He
D.Be

133. According to the general gas equation density of an ideal gas depends upon:
A.Pressure
B. Temperature
C. Molar mass of the gas
D.All of above

134. The actual volume of gas molecules is considered negligible at following pressures:
A.2 atm
B. 4 atm
C. 6 atm
D.8 atm

135. CO2 and SO2 both are triatomic molecules but heat of vaporization of SO2 is greater than that of CO2 due to:
A.High electronegativity of S
B. Greater size of SO2
C. SO2 is polar and CO2 is non-polar
D.SO2 is more acidic than CO2

136. Which of the following has the lowest vapour pressure at 20°C?
A.Diethyl ether
B. Chloroform
C. Carbon tetrachloride
D.Water

137. Which of the following is not a molecular solid?
A.Bromine
B. Sulphur
C. Phosphorus
D.Carbon dioxide

138. The lattice energy is also called as:
A.Energy of affinity
B. Bond energy
C. Crystal energy
D.Potential energy

139. For a gaseous phase reaction, when number of moles of reactants and products are equal:
A.The values of Kp and Kc are different
B. The values of Kp is greater than Kc
C. The values of Kc is greater than Kp
D.The values of Kp and Kc are the same

140. Purification of table salt (NaCl) by passing HCI gas through its saturated aqueous solution is an example of:
A.Law of mass action
B. Hess's law
C. Common ion effect
D.Henry's law

PHYSICS

141. If during circular motion tangential velocity of a body becomes double
then centripetal force becomes:
A.Double
B. One half
C. Four times
D.One fourth

142. Under what condition an object will have zero displacement but non zero
distance?
A.Linear motion
B. Circular motion
C. Random motion
D.Oscillation

143. Which one of the following properties is not exhibited by the longitudinal

MDCAT Past Papers
PRESTIGIOUS TESTING SERVICE
2000-2023

waves?
A.Interference
B. Reflection
C. Deflection
D.Polarization

144. The speed of sound in the air is 332 m/s. The speed of sound at 22°C will be:
A.345.2 m/s
B. 340 m/s
C. 350 m/s
D.330 m/s

145. Astronomers calculate speed of distant stars and galaxies using which of the following phenomena?
A.Beats
B. Interference
C. Superposition Principle
D.Doppler effect

146. In a ripple tank, 40 waves pass through a certain point in one second. If the wavelength of the wave is 5 cm, then speed of the wave is:
A.0.5 ms-1
B. 1 ms-1
C. 1.5 ms-1
D.2 ms-1

147. In which process the entire of heat supplied to the gas is converted to the internal energy of the gas?
A.Isochoric Process
B. Isobaric Process
C. Isothermal Process
D.Adiabatic Process

148. The internal energy of a system during an isothermal process:
A.Decrease
B. Increase
C. Becomes zero
D.Remain constant

149. If the potential at a point which is 1m from a charge is 1 volt, then the potential at a point which is 2m from the same charge will be:
A.2v
B. 1v
C. 0.5v
D.3v

150. The values of electric intensity will_____ due to the presence of dielectric medium.
A.Increase
B. Increase exponentially
C. Decrease
D.Remain same

151. The slope of distance time graph will always be:
A.Negative
B. Positive
C.0
D.Maximum

152. At what angle of projection of a projectile the range becomes half of its maximum value?
A.15°
B. 20°
C. 30°
D.40°

153. If we drop an object, its initial velocity is zero, how far will it fall in time "t"?
A.9.8 t2
B. 4.9 t2
C. 0.49 t2
D.98 t2

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

154. The Newton-second is unit of:
A.Work
B. Power
C. Impulse
D.Momentum

155. 1.75 m height weight-lifter rises weights with a mass of 50 kg to a height of
0.5 m above his head. How much work is being done by him? (g=10ms-²)
A.2125 J
B. 250 J
C. 100 J
D.50 J

156. What is the speed of 2.0 kg metallic bob at the mean position of a simple
pendulum, when released from its extreme position 0.5 m height? (g=10ms-²)
A.3.16 ms-1
B. 10 ms-1
C. 100 ms-1
D.50 ms-1

157. When the speed of your car is halved, by what factor does its kinetic energy
decreases?
A.1/2
B. 1/4
C. 1/8
D.1/6

158. Which one of the following force is non conservative force?
A.Frictional force
B. Gravitational force
C. Electric force
D.Elastic spring force

159. The Earth rotates on its Axis once a day. suppose, by some process earth contracts so that its radius is only half as large as present, then along the earth will take to complete it's rotation?
A.24 hours
B. 18 hours
C. 6 hours
D.12 hours

160. 1 Radian is equal to:
A.57.1°
B. 57.2°
C. 57.3°
D.57.4°

161. In translation from grid station, power losses are minimised by:
A.Increasing current
B. Decreasing current
C. Increasing resistance
D.increasing voltage

162. The domestic electricity supply has a frequency of:
A.150 Hz
B. 100 Hz
C. 50 Hz
D.25 Hz

163. PIV stands for:
A.Positive Inverse voltage
B. Power integrated voltage
C. Peak inverse voltage
D.Peek integrated voltage

164. In full wave rectification, the diodes are used:
A.1
B. 2
C. 3
D.4

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

165. The wavelength associated with an electron is of the order of:
A.Visible light
B. X - rays
C. Radio waves
D.Infrared

166. Which Photon carries the most energy?
A.Blue
B. Voilet
C. Red
D.Green

167. Which one of the following series lies in the ultraviolet region?
A.Balmer series
B. Paschen series
C. Lyman series
D.Bracket series

168. The main difference between X-rays and γ rays is?
A.Frequency
B. Wavelength
C. Energy
D.Origin

169. There are initially 400 atoms in a radioactive sample. what would be the number of atoms after three half-life?
A.400
B. 200
C. 50
D.25

170. While using radiation therapy, cancerous thyroid is treated with

_____
radioisotope:
A.Carbon
B. 235-Uranium
C. Thorium
D.Iodine-131

171. In capacitors, energy is stored in the form of:
A.Gravitation energy
B. Kinetic energy
C. Electric intensity
D.Magnetic induction

172. Ohm time's farads is equivalent to:
A.Time
B. Charge
C. Distance
D.Capacitors

173. One-kilowatt hour is commonly termed as one commercial unit of electric
energy which is equal to:
A.3.6 × 105 J
B. 3.6 × 106 J
C. 3.6 × 104 J
D.3.6 × 103 J

174. When a wire is compressed and its radius become 2R then its resistance will be:
A.16 R
B.4 R
C. 1/16 R
D.1/4 R

175. One of the following is an ohmic device:
A.Filament bulb
B. Semiconductor diode
C. Tranistors
D.Copper wire

176. The change in resistance of metallic conductor at temperature below 0°C is:
A.Non linear
B. Curve
C. Linear
D.Curvilinear

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

177. When current are flowing through two long parallel wires same direction electric field between them:
A.Strong
B. Weak
C. Remains same
D.Infinite

178. Magnetic flux is maximum when angle between magnetic field and vector
area is:
A.0°C
B. 90°C
C. 180°C
D.45°C

179. Transformer is a device which steps up or steps down the input:
A. Current
B. Voltage
C. Energy
D. Power

180. If a stationary bar magnet is placed near a coil at rest maximum lines of
force pass through the coil, the Galvanometer shows:
A. Maximum current
B. Minimum current
C. No current
D. Intermediate value of current

ENGLISH

181. We prefer fruits _____ sweets.
A.To
B. On
C. Over
D.From

182. Choose the correct spelling:
A.Exantuated
B. Axantualed
C. Accenchuated
D.Accentuated

183. Choose the correct spelling:
A.Cotioned
B. Cautioned
C. Causchuned
D.Coschuned

184. Choose the correct spelling:
A. Eccentric
B. Eccentrac
C. Akcantric
D. Accentric

185. Choose the correct spelling:
A. Dafinite
B. Defanite
C. Dafanite
D. Definite

186. The Headmaster_____ to speak to you.
A.Wants
B. Is wanting
C. Was wanting
D.Want

187. Choose the correct option:
Knowledge and wisdom _____ no time for connection.
A. Has
B. Have
C. Had
D. Are

188. Identify the errors and choose the correct option:
A.I hope this letter will find you in good of high spirits
B. I hope this letter finds you in best of your spirit
C. I hope letter finds you in the best of spirits

MDCAT Past Papers BY PRESTIGIOUS TESTING SERVICE 2000-2023

D.I hope the letter found you in greatest of spirite

189. Identify the errors and choose the correct option:
A. Gulliver travel was written by swift
B. Gulliver travels was written to swift
C. Gulliver's travels was written by swift
D.Gulliver's travel was written by swift

190. Fill in the blank with the appropriate article as required. Umbrella is of no avail against a thunderstorm.
A.The
B. A
C. An
D.No article required

191. Choose the correct sentence:
A.I wish I have been a millionaire
B. I wish I am being a millionaire
C. I wish I were a millionaire
D. I wish I was millionaire

192. Pick the correct option:
A.No star is brighter than the moon
B. No star is more bright than the moon
C. No star is brighter then the moon
D.No star is brighter than moon

193. Choose the correctly structured sentence:
A.Had he lived in England he would miss his family
B. Had he lived in England, he would have missed his family
C. Had he lives in England he had missed his family sin England
D.Had he live in England he will missed his family

194. She always carried an umbrella. The sentence indicates tense.
A.Present tense
B. Past Simple
C. Past perfect
D.Present perfect

195. Ahmed _____ me for a long time.
A.Know
B. Have known
C. Knows
D.Knew

196. Pick the correct option:
A.His first inning's consists of four 6 and three 4
B. His first innings consist of four 6's and three 4's
C. His first innings consist of four 6's and three 4's
D.His first inning's consist of four 6's and three 4's

197. Choose the correctly punctuated sentence:
A.What a fall was there, my countrymen! Long live the king!
B. What a fall was there! My countrymen. Long live the King!
C. What a fall was there, my countrymen, Long live the king.
D.What a fall was there, my countrymen, Long live the king

198. Choose the correct option:
A.He and I was playing
B. He and I were playing
C. He and I were being playing
D.He and I was being playing

199. Choose the correct option:
A.Every one of one of the prisons are full
B. Every one of the prisons had full
C. Every one of the prisons have full
D.Every one of the prisons is full

200. Not only the parents but also their _____ son for interview.

MDCAT Past Papers
PRESTIGIOUS TESTING SERVICE
2000-2023

A.Has called
B. Have called
C. Have been called
D.Has been called

MDCAT
Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

ANSWER KEY

Q NO	ANS KEY	Q NO	ANS KEY	Q NO	ANS KEY	NO	ANS KEY
1	B	26	B	51	D	76	C
2	C	27	D	52	A	77	A
3	B	28	C	53	B	78	C
4	D	29	D	54	D	79	D
5	A	30	A	55	B	80	D
6	C	31	A	56	B	81	B
7	B	32	B	57	D	82	C
8	A	33	C	58	A	83	B
9	D	34	B	59	D	84	B
10	A	35	B	60	D	85	C
11	A	36	A	61	C	86	C
12	A	37	A	62	A	87	D
13	B	38	B	63	D	88	C
14	B	39	C	64	C	89	B
15	B	40	B	65	C	90	C
16	C	41	A	66	C	91	B
17	E	42	C	67	D	92	D
18	A	43	B	68	C	93	D
19	D	44	D	69	D	94	A
20	C	45	B	70	A	95	B
21	D	46	B	71	B	96	B
22	D	47	C	72	B	97	B
23	A	48	B	73	C	98	B
24	C	49	B	74	C	99	A
25	D	50	B	75	B	100	C

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023

ANSWER KEY

Q NO	ANS KEY	Q NO	ANS KEY	Q NO	ANS KEY	NO	ANS KEY
101	C	126	C	151	B	176	C
102	D	127	C	152	A	177	B
103	A	128	C	153	C	178	A
104	B	129	C	154	C	179	B
105	B	130	A	155	B	180	C
106	C	131	B	156	A	181	A
107	B	132	D	157	B	182	D
108	B	133	D	158	A	183	B
109	A	134	A	159	C	184	A
110	B	135	C	160	C	185	D
111	C	136	D	161	D	186	A
112	B	137	A	162	D	187	B
113	A	138	B	163	C	188	A
114	D	139	D	164	B	189	C
115	D	140	C	165	B	190	C
116	C	141	C	166	B	191	C
117	A	142	B	167	C	192	A
118	C	143	D	168	D	193	B
119	B	144	A	169	C	194	B
120	D	145	D	170	D	195	D
121	A	146	D	171	C	196	C
122	B	147	A	172	A	197	A
123	D	148	D	173	B	198	B
124	C	149	C	174	D	199	D
125	B	150	C	175	D	200	D

MDCAT Past Papers
BY PRESTIGIOUS TESTING SERVICE
2000-2023